nature plants

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https://doi.org/10.1038/s41477-020-0670-y

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Multiplexed heritable gene editing using RNA viruses and mobile single guide RNAs

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An in planta gene editing approach was developed wherein Cas9 transgenic plants are infected with an RNA virus that expresses single guide RNAs (sgRNAs). The sgRNAs are augmented with sequences that promote cell-to-cell mobility. Mutant progeny are recovered in the next generation at frequencies ranging from 65 to 100%; up to 30% of progeny derived from plants infected with a virus expressing three sgRNAs have mutations in all three targeted loci.

Gene editing with sequence-specific nucleases offers great promise for advancing basic and applied plant research[1]. Realizing this promise, however, is not without challenges. For example, nearly all heritable mutations created through gene editing rely on the use of tissue culture, which has many drawbacks: it works with limited species and genotypes, requires considerable time and may cause unintended changes to the genome and epigenome[2]. To avoid tissue culture, attempts have been made to achieve transformation by delivering transgenes to meristems or egg cells; however, this has really been successful only with *Arabidopsis thaliana* and its close relatives[3,4]. Clearly, there is a need for novel, alternative approaches to deliver gene editing reagents to create edited plants more efficiently.

Ribonucleic acid viruses have largely been unexploited as vectors for plant genetic modification, the exception being their use for viral-induced gene silencing (VIGS) wherein a fragment of a host gene expressed by a viral vector triggers silencing of the gene in the host plant[5]. As vectors, RNA viruses are handicapped by limited cargo capacity, which is typically <1 kb, precluding their use for delivery of gene editing reagents such as Cas9 from *Streptococcus pyogenes* (4.1 kb). Several groups, however, have used RNA viruses to deliver sgRNAs to transgenic plants expressing Cas9 (refs. [6–9]). This approach typically results in low frequencies of gene editing in somatic cells, and recovery of mutant progeny is rare. Heritable editing events were recovered in only one study, and at a frequency of 1/438 seedlings[6].

We reasoned that heritable gene edits could be generated at higher frequencies using RNA viral vectors if the sgRNA could gain better access to the germline. In plants, there are many examples of endogenous mobile RNA sequences, one of which is encoded by *Flowering Locus T* (*FT*)[10,11]. FT is transcribed in leaf vascular tissue and then moves to the shoot apical meristem to induce flowering[11]. FT RNA has previously been demonstrated to enable cell-to-cell movement when fused to a messenger RNA that binds green fluorescent protein (GFP)[12]. We hypothesized that, if fused to a sgRNA, FT might promote entry into the shoot apical meristem and create heritable gene edits at higher frequencies (Fig. 1a,b). To test this hypothesis, we added the *A. thaliana* FT coding sequence to the 3′-end of a sgRNA that targets all alleles of *phytoene desaturase* (PDS) in *Nicotiana benthamiana* (Fig. 1a and Supplementary Table 6). Loss of PDS function results in photo-bleaching due to a lack of photoprotective carotenoids[13]. We also generated sgRNAs fused to a mutant FT (bearing a mutation in the start codon, mFT) and a truncated FT, both of which have been shown to promote mobility of GFP-encoding RNAs to the meristem[14] (Supplementary Table 6).

The sgRNAs fused to FT sequences were cloned into a tobacco rattle virus (TRV) vector that can be delivered to plants via infection by *Agrobacterium tumefaciens*[15] (Fig. 1a,b). *A. tumefaciens* strains with TRV vectors were infiltrated into leaves of a transgenic line of *N. benthamiana* that expresses *Sp*Cas9 (ref. [16]). To determine whether the addition of FT sequences affected sgRNA function, DNA was prepared from the infiltrated tissue and assessed for mutations at PDS. Frequencies of mutagenesis were comparable between those sgRNAs with FT sequences and a control (Extended Data Fig. 1 and Supplementary Table 1a). Approximately 2.5 weeks after infiltration, bleached spots began to appear on the upper systemic leaves of infected plants (Fig. 1c). This phenotype became more prominent in newly emerging leaves as the plant matured. In several cases, the entire upper growing portion of the plant was completely white (Fig. 1d). Next-generation sequencing (NGS) suggested that the bleached phenotype was due to loss of PDS function; mutagenesis frequencies in the eighth systemically infected leaf exceeded 80% (Fig. 1e and Supplementary Table 1b). Vectors with unmodified sgRNAs also resulted in gene editing in systemic leaves, as previously observed[6], although at a lower average frequency and with greater variability.

We next tested whether the high editing efficiency observed in systemic leaves resulted in germline modifications that were transmitted to seedlings. Parental plants infected with TRV vectors expressing mobile sgRNAs produced some fully bleached seedlings (Fig. 1h), which would be expected if germline mutations occurred at all PDS alleles. This was in fact the case, as genotyping of seedlings derived from infected parents indicated that up to 65% of seedlings contained a mutation in at least one PDS allele (Fig. 1f and Supplementary Table 3). The effectiveness of FT in increasing mutagenesis was also evident by the abundance of heritable heterozygous and bi-allelic mutations when FT was added (Fig. 1g). There was no significant difference in editing efficiency between the functional FT, mFT and truncated FT constructs tested, and no effect was observed on flowering time. High editing efficiency with

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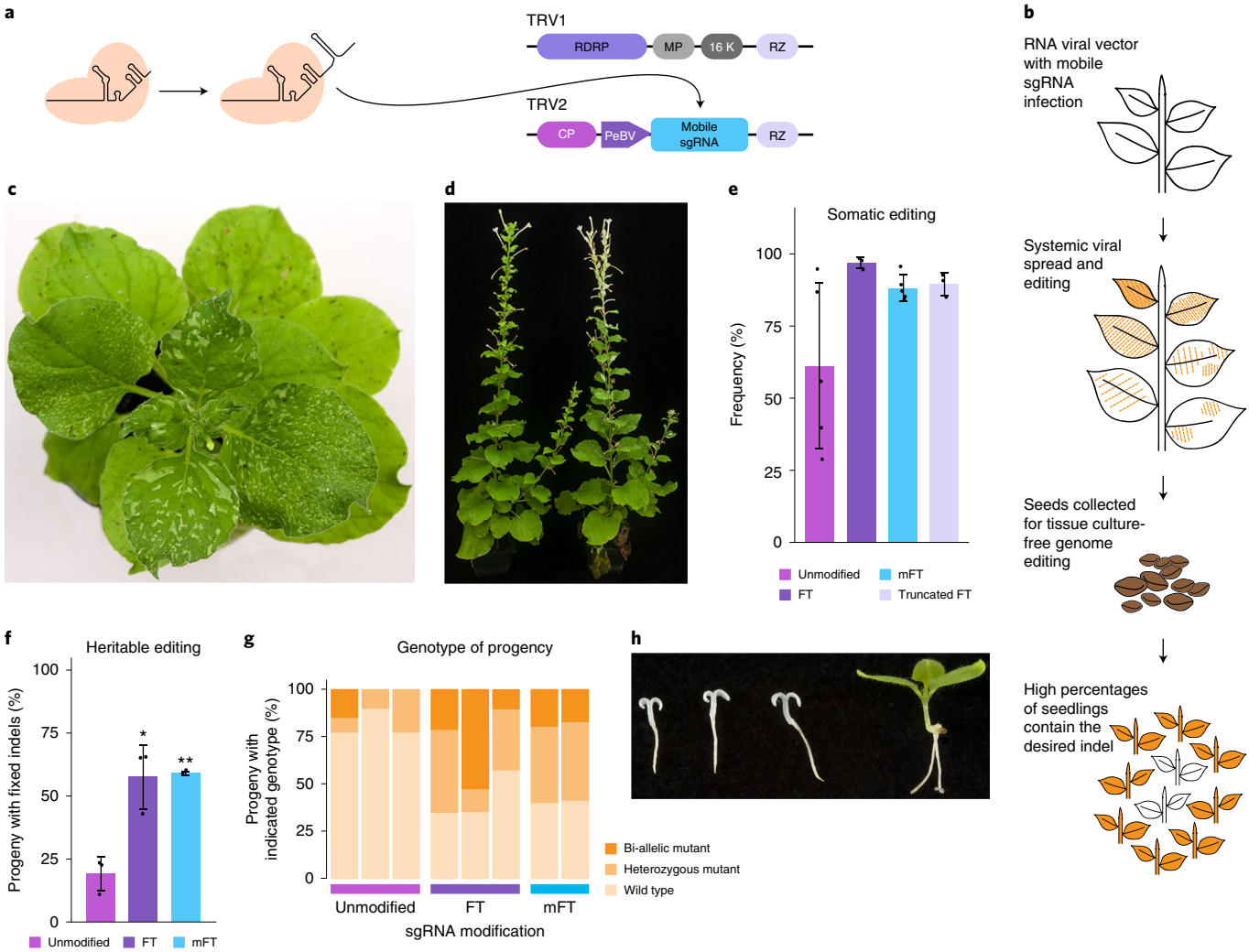


Fig. 1 | Heritable gene editing using RNA viruses and mobile sgRNAs. a, sgRNAs were modified by the 3′ addition of RNA sequences that promote mobility. Many of these mobility motifs contain secondary structures, hypothesized to be important for function. Modified sgRNAs were cloned into a TRV vector for replication and systemic plant infection. RDRP, RNA-dependent RNA polymerase; MP, movement protein; CP, coat protein; PeBV, pea early browning virus subgenomic promoter; RZ, terminating ribozyme. **b**, Overview of the method for creating heritable gene edits. RNA viral vectors expressing a modified sgRNA were used to infect Cas9-expressing plants; gene editing occurred as the virus moved systemically throughout the plant. After the plant matures, seed is collected and progeny have mutations in the desired locus. **c**, Phenotype of *N. benthamiana* plants infected with TRV expressing a modified sgRNA targeting PDS. The plant shown was photographed approximately 4 weeks after agro-infiltration. White blemishes are indicative of loss of PDS function. The observed phenotype appears approximately 2.5 weeks after agro-infiltration and increases in intensity as the plant matures. **d**, Phenotype of infected *N. benthamiana* plants approximately 10 weeks after agro-infiltration. Left and right photographs show a plant not infected and a plant infected with TRV expressing modified sgRNAs targeting PDS, respectively: note the white blemishes on the lower leaves and that the upper portion of the plant is completely white. White areas were observed on all 11 independent biological replicates. **e**, Editing efficiency of mobile sgRNAs targeting PDS at the eighth systemic leaf (eight leaves above the infiltrated leaf). A higher average editing frequency and lower variance were observed for FT ($n=3$ independent biological replicates, $\bar{x}=0.97\pm0.02$), mFT ($n=5$, $\bar{x}=0.88\pm0.05$) or truncated FT ($n=3$, $\bar{x}=0.90\pm0.04$) compared to unmodified sgRNA ($n=5$, $\bar{x}=0.61\pm0.29$). The higher variability from the unmodified sgRNA is probably due to inconsistent viral movement. Editing efficiency was calculated by NGS of PCR amplicons spanning the targeted locus; frequency equals the fraction of reads, with indels divided by the total number of reads. Each dot represents one plant replicate. Bars indicate the mean editing frequency, with error bars representing ±s.d. **f**, Heritable gene editing from RNA viral vectors. sgRNAs modified with FT ($n=3$ independent biological replicates, $\bar{x}=0.58\pm0.13$, $P=0.018$) or mFT ($n=2$, $\bar{x}=0.59\pm0.01$, $P=0.009$) produced significantly higher frequencies of heritable editing compared to an unmodified sgRNA ($n=3$, $\bar{x}=0.19\pm0.07$). Up to 65% of progeny genotyped contained an indel mutation. Seeds were collected from the entire plant and pooled. Heritable editing frequency is the fraction of progeny containing an indel in at least one allele, divided by the total number of progeny genotyped. Each dot represents the heritable editing frequency in progeny of one parent replicate infected with TRV expressing sgRNAs with FT ($n=23$, 17 or 28 progeny), mFT ($n=10$ or 17) or no modification ($n=13$, 19 or 13). Bars indicate the mean editing frequency, with error bars representing ±s.d. Asterisks indicate statistically significant differences compared with the unmodified sgRNA, by two-sided Student's *t*-test, $^{*}P<0.05$, $^{**}P<0.01$. **g**, Mobile sgRNAs produce a higher portion of bi-allelic and heterozygous mutations compared to unmodified sgRNAs. Heterozygous mutant progeny contain one wild-type allele and one mutant allele; bi-allelic mutant progeny contain mutations in both alleles. The percentage of each genotype was determined by the fraction of progeny containing that genotype divided by total progeny assessed. The same progeny used to determine heritable editing frequency in **f** were used to calculate genotype percentage. **h**, PDS knockout phenotype in *N. benthamiana* seedlings. Several seedlings from parents infected with TRV expressing a modified sgRNA targeting PDS were completely white, indicating a heritable complete knockout of PDS and validation of genotyping results. When extracting DNA for genotyping, white seedlings were chosen in proportion to the number of white versus green seedlings. White seedlings were observed from all five independent biological replicates that had been infected with TRV expressing modified sgRNAs.

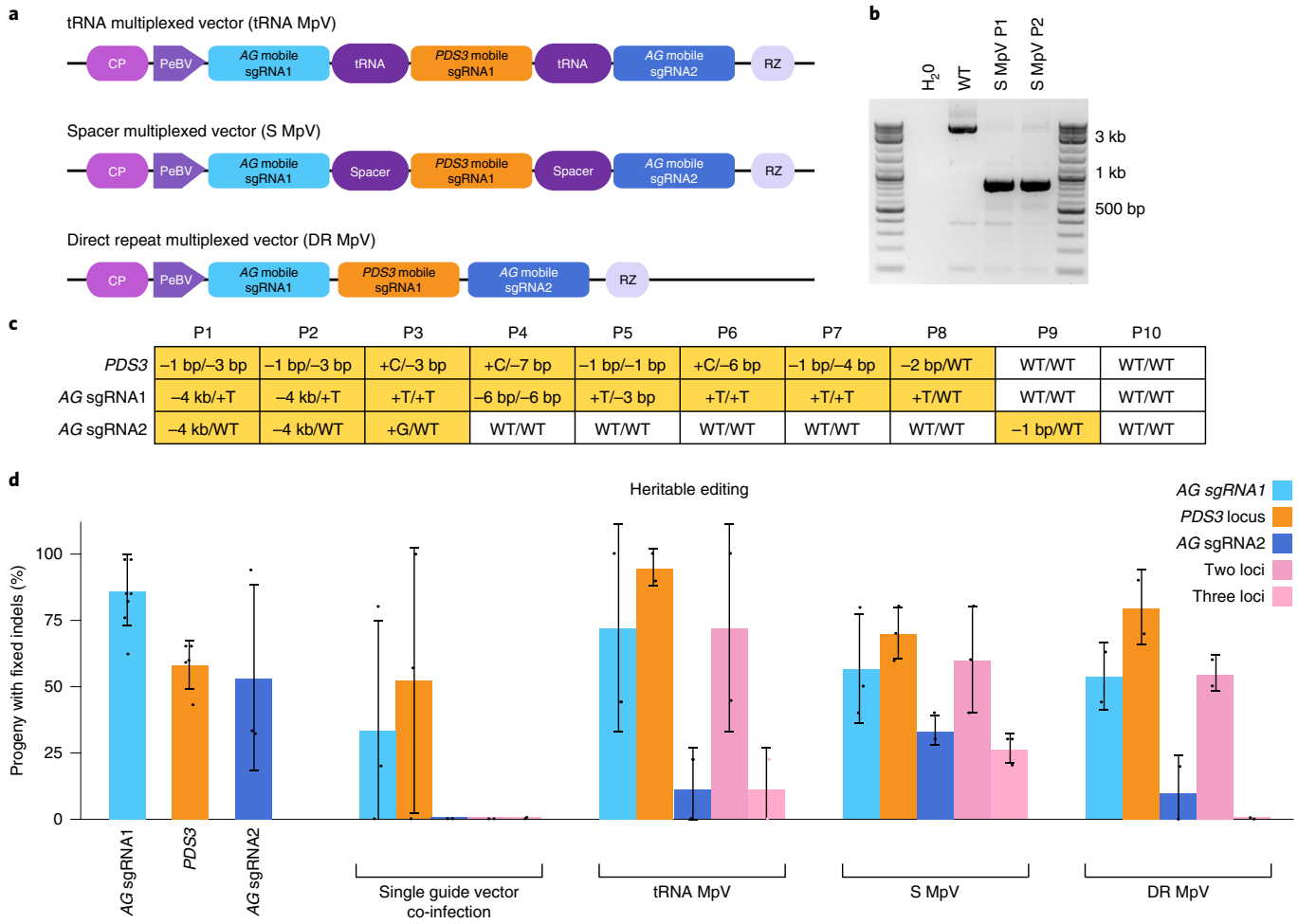


Fig. 2 | Multiplexed heritable gene editing using an RNA virus and mobile sgRNAs. a, Three TRV2 vectors were tested for their ability to create heritable mutations at multiple loci. Each vector expressed three sgRNAs (*PDS3*, *AG* sgRNA1 and *AG* sgRNA2); all sgRNAs were fused to the truncated FT motif to enhance mobility. The vectors differed in terms of sequences separating the sgRNAs: a tRNA (vector tRNA multiplexed vector (MpV)), a 23-bp spacer sequence (vector S MpV) and a 4-bp cloning site (vector DR MpV). **b**, Some progeny of plants infected with the spacer multiplexed vectors sustained a 4.05-kb deletion resulting from loss of DNA between the *AG* sgRNA1 and *AG* sgRNA2 target sites. The gel shows PCR amplification products from WT and two independent mutant progeny from one parent infected with the spacer multiplexed vector next to a 1-kb plus DNA ladder. **c**, Ten progeny were genotyped from a *N. benthamiana* parent infected with the spacer MpV. Both bi-allelic and multiplexed mutations were recovered. The target locus is indicated on the left, while P1, P2 and so on indicate individual progeny genotyped. (–) indicates a base pair deletion; inserted bases are preceded with a (+) and the nucleotide inserted. **d**, Multiplexed RNA viral vectors produce heritable modifications in multiple loci. The editing frequency observed at each targeted locus with multiplexed vectors is similar in frequency to that observed with vectors expressing single guides (shown on the left; *AG* sgRNA1 ($n=8$ independent biological replicate parents, $\bar{x}=0.86\pm0.14$), *PDS3* ($n=5$ parents, $\bar{x}=0.58\pm0.09$) or *AG* sgRNA2 ($n=3$ parents, $\bar{x}=0.53\pm0.35$). Single sgRNA vectors co-infected into the same plant do not reliably produce heritable edits. Large error bars are the result of inconsistencies, with one parent producing high frequencies of editing at one locus and very low frequencies at another. No progeny with multiplexed gene edits from co-infected vectors was observed. Co-infection: $n=3$ independent biological replicate parents (*AG* sgRNA1, 0.33 ± 0.42; *PDS3*, $\bar{x}=0.52\pm0.50$; *AG* sgRNA2, $\bar{x}=0\pm0$; two loci, $\bar{x}=0\pm0$; three loci, $\bar{x}=0\pm0$). Multiplexed heritable editing was observed with all three vector designs tested. tRNA multiplexed, $n=2$ independent biological replicates (*AG* sgRNA1, $\bar{x}=0.72\pm0.39$; *PDS3*, $\bar{x}=0.95\pm0.07$; *AG* sgRNA2, $\bar{x}=0.11\pm0.16$; two loci, $\bar{x}=0.72\pm0.39$; three loci, $\bar{x}=0.11\pm0.16$). Spacer multiplexed, $n=3$ independent biological replicates (*AG* sgRNA1, $\bar{x}=0.57\pm0.21$; *PDS3*, $\bar{x}=0.70\pm0.10$; *AG* sgRNA2, $\bar{x}=0.33\pm0.06$; two loci, $\bar{x}=0.60\pm0.20$; three loci, $\bar{x}=0.27\pm0.06$). Direct-repeat multiplexed, $n=2$ independent biological replicates (*AG* sgRNA1, $\bar{x}=0.53\pm0.13$; *PDS3*, $\bar{x}=0.80\pm0.14$; *AG* sgRNA2, $\bar{x}=0.10\pm0.14$; two loci, $\bar{x}=0.55\pm0.07$; three loci, $\bar{x}=0\pm0$). Heritable editing at each locus is the ratio of progeny with modifications at that locus to the total number of progeny with that locus genotyped. Heritable editing frequency in two loci is the ratio of progeny with modifications in at least two loci to the total number of progeny with at least two loci genotyped. Heritable editing frequency in three loci is the ratio of progeny with modifications in three loci to the total number of progeny with all three loci genotyped. Each dot represents the heritable editing frequency of sgRNA progeny from one parent replicate. Bars indicate the mean editing frequency, with error bars representing ±s.d.

the modified sgRNAs was not a property unique to the PDS locus. Similar frequencies of somatic editing were observed at a second locus, namely *AGAMOUS* (AG sgRNA1), and mutations in AG were transmitted to the next generation at frequencies as high as 100% (Extended Data Fig. 2 and Supplementary Tables 2 and 4). These mutations were fixed and transmitted for two generations (Extended Data Fig. 3 and Supplementary Table 5).

Flowering Locus T is not the only RNA that moves between cells. Plant phloem exudates contain many transfer RNA-like sequences capable of cell-to-cell movement[17]. We tested whether methionine,

glycine and isoleucine tRNAs could improve editing efficiency when individually fused to the 3′-end of the PDS sgRNA. As observed with FT, infected plants became photo-bleached (Extended Data Fig. 4) and produced some fully photo-bleached seedlings (Extended Data Fig. 5b). Several bleached or green seedlings were genotyped and confirmed to have mutations in PDS (Extended Data Fig. 6). We performed quantitative PCR with reverse transcription (RT–qPCR) on RNA extracted from systemically infected leaves, and found a higher amount of viral RNA produced from TRV vectors with sgRNAs augmented with FT, truncated FT or tRNAs relative to vectors lacking augmented sgRNAs (Extended Data Fig. 5a). This suggests that RNA mobility sequences increase overall virus accumulation which, in turn, results in higher frequencies of heritable gene editing. The increase in viral abundance is transient, however, as we did not detect any evidence of viruses in progeny (Extended Data Fig. 7).

One advantage of the CRISPR/Cas9 system is that several loci can be targeted at once by simultaneous expression of multiple sgRNAs (multiplexing). Multiple, functional sgRNAs have previously been expressed from viruses[7], and we designed three vector architectures expressing three mobile sgRNAs augmented with truncated FT. The sgRNAs target PDS and two sites in AG. The individual sgRNAs were either arranged in tandem or separated by either a 23-base pair (bp) spacer or a tRNA, the latter being processed from the RNA, resulting in release of the three sgRNAs[18] (Fig. 2a). A variety of editing outcomes were observed in the progeny of plants infected with the three vector architectures. Vectors with the 23-bp spacer were capable of inducing a 4.05-kb deletion between the two AG sgRNA target sites in multiple progeny (Fig. 2b). When single sgRNA vectors were co-infected, no progeny had edits in more than one target, consistent with reports of TRV superinfection exclusion[19]. Heritable editing efficiencies from multiplexed vectors at each individual locus were comparable to the delivery of single sgRNAs per se, and we were frequently able to recover plants with mutations at two or three targeted loci (Fig. 2b–d and Supplementary Table 7).

Although several groups have used RNA viral vectors to express sgRNAs for plant gene editing[6–9], only one report documented the generation of heritable mutations and then only at a very low frequency[6]. We observed several examples of non-augmented sgRNAs that resulted in lower, but nonetheless readily detectable, frequencies of heritable gene editing (Fig. 1f and Extended Data Fig. 2). A notable difference between our experiments and those previously performed is that we pooled seeds from the entire plant, including pods distal to the point of infiltration. Seed pods that develop later have a much higher fraction of progeny with mutations (Extended Data Fig. 8). In a previous report of low-frequency mutagenesis, only the first pods were analysed[6]. Growth conditions or parameters that affect the extent of virus infection may also impact mutagenesis frequency. We observed an impact of growth conditions on heritable editing frequency (Extended Data Fig. 9); however, a systematic study is required to determine specific environmental factors that may promote higher editing frequencies.

We report a method to achieve highly efficient, multiplexed, heritable gene editing using RNA viral vectors. The method reduces the need for tissue culture—a current bottleneck in plant gene editing. Only an initial Cas9 transgenic plant needs to be generated, which can be used repeatedly to create almost unlimited genetic diversity. Furthermore, the method could be used to create precise DNA sequence changes by infecting transgenic plants expressing other gene editing reagents such as base editors. Because TRV is widely used as a VIGS vector for gene function studies in both crop and model plants[5], TRV vectors expressing mobile sgRNAs can probably be used to create heritable gene edits in diverse plant species, helping to realize the promise of plant gene editing for advancement of basic and applied plant research.

Methods

Vector construction. Schematics of vectors to clone augmented sgRNAs are shown in Extended Data Fig. 10. The FT mobility sequence was isolated by first extracting RNA from the Columbia ecotype of *A. thaliana* using a TRIzol RNA extraction protocol (Invitrogen). The FT mRNA sequence was PCR amplified from the isolated RNA using a PCR with a reverse transcription (RT–PCR) kit (QIAGEN) and primers PP331 and PP332 (all primers are listed in Supplementary Table 6). The FT sequence was added to the 3′ end of a sgRNA targeting the PDS locus of *N. benthamiana* (*PDS*, Niben101Scf14708g00023.1). This was accomplished by PCR amplification of the sgRNA sequence using primers EE562 and EE267, and PCR amplification of the FT cDNA using primers EE271 and EE272. All PCR reactions for cloning were performed with Q5 DNA polymerase (New England Biolabs). The resulting PCR products were then cloned into the TRV2 cloning vector, pEE083 (vector sequences have been uploaded to Addgene). Cloning was accomplished by the addition of 0.5 µl of each PCR product to a mixture of 1 µl of pEE083 (50 ng), 0.5 µl of AarI restriction enzyme (Thermo Fisher Scientific), 0.4 µl of AarI oligonucleotide (Thermo Fisher Scientific), 1 µl of T4 DNA ligase (New England Biolabs), 2 µl of T4 DNA ligase buffer (New England Biolabs) and 14.1 µl of distilled H_2O. The reaction mixture was then cycled 10× (37 °C/5 min + 16 °C/10 min) plus 37 °C/10 min plus 80 °C/5 min, to create pEE391. mFT and truncated FT vectors were added to the 3′ end of the sgRNA by amplifying the FT complementary DNA using primers EE273 and EE272 (mFT) or EE273 and EE274 (truncated FT). These PCR products were combined with the previously amplified sgRNA and added to the pEE083 cloning vector in the same manner, to create pEE392 (mFT sgRNA) or pEE393 (truncated FT sgRNA). Unmodified control sgRNA vectors were created by PCR amplification of the sgRNA sequence using primers EE562 and EE282 and assembling the amplification products into pEE083 to create pEE390.

Vectors targeting *AGAMOUS* (AG, Niben101Scf12205g00011.1) were assembled in the same manner as the PDS vectors, with the exception that primer EE659 was used instead of EE562 to amplify sgRNA. This created AG-targeting vectors pEE386 (unmodified sgRNA control), pEE387 (FT sgRNA), pEE388 (mFT sgRNA) or pEE389 (truncated FT sgRNA). To facilitate rapid assembly of unique modified sgRNA sequences, we developed a cloning vector that uses one unique forward primer (containing the sgRNA target of interest), one constant reverse primer and one plasmid for PCR amplification and assembly of the vector (pEE515; Extended Data Fig. 10). We recommend using the truncated mFT to reduce the potential effects of early or delayed flowering, even though no differences in flowering were observed in *N. benthamiana*.

Spacer and direct-repeat multiplexed vectors were assembled, by PCR amplification with Q5 DNA polymerase, the mobile sgRNAs from the pEE515 template using forward primers EE659 (AG sgRNA1), EE884 (PDS) or EE885 (AG sgRNA2) and reverse primers EE886, EE867, EE274 (respectively, spacer) or EE765,EE766,EE274 (respectively, direct repeat). Each PCR amplicon (0.5 µl) was combined and cloned into pEE515 using the same protocol listed above to create pEE531 (spacer) or pEE495 (direct repeat). tRNA multiplexed vectors were created by first assembling a vector with a single guide by PCR amplification of the mobile sgRNA using EE659 and EE733, with pEE515 as a template, and tRNA using EE734 and EE735 with pMOD_B2303 (ref. [20]) as a template. Each PCR amplicon (0.5 µl) was combined and cloned into pEE515. This template was then used for PCR amplification of each mobile sgRNA using EE659 and EE736, EE884 and EE738, and EE885 and EE274. PCR amplicons were combined and cloned into pEE515 to create pEE491.

To generate vector SPDK3876 (TRV-RNA2-pPEBV-MCS), the pea early browning virus (PEBV) promoter and a multiple cloning site were PCR amplified using SP9008 and SP5411 and cloned into pYL156 (ref. [15]) that had been digested with AatII and XmaI. To generate vectors SPDK3888 (TRV-RNA2-pPEBV-tRNA^{Ile}), SPDK3889 (TRV-RNA2-pPEBV-tRNA^{Met}) and SPDK3902 (TRV-RNA2-pPEBV-tRNA^{Gly}), the annealed primer pairs SP8781 + 8782, SP8783 + 8784 and SP8779 + 8780, corresponding to different tRNA sequences, were cloned into SPDK3876 digested with SacI and XmaI. The sgRNA-targeting PDS was added by annealing SP8790 + 8791 and 8788 + 8789 and cloning the fragments into vectors 3888, 3889 and 3902 digested with XbaI and SacI.

Infection of *N. benthamiana* and assay of somatic cell-editing efficiency. *Agrpbacterium tumefaciens* strain GV3101 was used to deliver all vectors into *N. benthamiana* (GenBank: PRJNA170566). Cultures containing the vector of interest were grown in lysogeny broth (LB) medium overnight and prepared for leaf infiltration by centrifugation for 10 min at 2,500 relative centrifugal force (RCF). The LB medium was discarded, and *A. tumefaciens* cells were resuspended in 4 ml of agro-infiltration buffer (10 mM $MgCl_2$, 10 mM 2-(*N*-morpholino) ethanesulfonic acid, pH 5.6). This suspension was again placed in a centrifuge for 10 min at 2,500 RCF and the agro-infiltration buffer discarded. The pelleted cells were resuspended in agro-infiltration buffer to $OD_{600} = 0.6$. At this point each TRV2 vector was mixed in a 1:1 ratio with the TRV1 vector pNJB069, and incubated at room temperature for 3 h. After incubation the resuspended cells were infiltrated into 6-week-old *N. benthamiana* plants overexpressing Cas9 at the underside of the fifth true leaf, using a needle-less syringe[21]. *N. benthamiana* plants were grown in a growth chamber (Conviron) under 26/22-°C day/night condition with a 12/12-h day/night light cycle, a light intensity of 100 µmol m^{-2} and 50% humidity.

Deoxyribonucleic acid was extracted from the infiltration site of *N. benthamiana* leaves 2 weeks after infiltration, to assay the ability of modified sgRNA to cleave DNA. DNA was extracted from the eighth systemic leaf approximately 6 weeks after infiltration, to assay the ability of modified sgRNA to enhance somatic cell editing through systemic infection. Two separate 8-mm leaf punches were extracted from the infiltrated site or on either side of the eighth systemic leaf midrib, approximately 4 mm from the base of the leaf. These punches were combined and used for genomic DNA extraction using a cetyltrimethylammonium bromide (CTAB) protocol[22]. To quantify editing efficiency, a PCR reaction was performed using this genomic DNA as a template, primers RN468 and RN473 (PDS) or EE693 and EE697 (AG) and a PHIRE Plant Direct PCR Kit (Thermo Fisher Scientific). The resulting amplicons were purified using a Gel Purification Kit (QIAGEN) and sent for Amplicon-EZ (Genewiz) NGS. Paired-end .fastq files resulting from NGS were analysed using CRISPR RGEN Tools[23] with a 40-bp comparison range to determine indel frequency.

Assay of fixed indels in infected *N. benthamiana* progeny. After a period of approximately 2 months post infiltration, or when at least eight seed pods had developed and opened, seed was collected from infected plants and all pods from a given plant were pooled. Seeds were sterilized with 50% bleach, placed on ½ Murashige and Skoog medium + 50 ng µl^{-1} kanamycin plates and maintained at 25 °C with a 16/8-h day/night cycle. After growth to the 4–6-leaf stage, DNA was extracted using a CTAB protocol[22]. Each purified DNA sample, representing one seedling, was used as a template for a PHIRE Plant Direct (Thermo Fisher Scientific) or Q5 DNA polymerase PCR reaction. Primers EE552 and EE504_R, EE653 and EE655, or EE628 and EE631 were used to amplify the PDS, AG sgRNA1 or AG sgRNA2 locus, respectively. Large deletions in the AG locus were detected by PCR amplification of the locus with oEE628 and oEE655. These PCR amplicons were purified using a PCR Purification Kit (QIAGEN) or an ExoSAP-IT (Thermo Fisher Scientific) and sent for Sanger sequencing. ICE software v.2.0 (Synthego) was used to quantify indel frequency. Due to noise in ICE-based indel detection, we used the following parameters to call genotypes: 0–10% indels, WT; 35–65% indels, heterozygous mutation (WT/indel); 85–100% indels, bi-allelic mutation. All seedlings assayed fell within these ranges.

Quantitative real-time PCR analysis. Total RNA was extracted using TRIzol reagent (Life Technologies) and treated with DNase I RNase free (New England Biolabs). First-strand cDNA was synthesized used SuperScript III reverse transcriptase (Life Technologies) and TRV-specific primer SP9239; the control was the NbPP2A-specific primer SP8061. RT–qPCR was performed using the Bio-Rad CFX96 qPCR instrument (Bio-Rad) with iTaq Universal SYBR Green Supermix (Bio-Rad), 2.5 µM each of TRV-specific primers (SP9238 + 9239) or NbPP2a-specific primers (SP8060 + 8061) and 5% diluted first-strand cDNA. Data presented were normalized for NbPP2A expression level.

RT–PCR. Total RNA was extracted as described for the RT–qPCR analysis. RT–PCR was performed on a subset of the samples by carrying out first-strand cDNA synthesis in the same manner as the RT–qPCR analysis. First-strand cDNA (1 µl) was used in a 20-µl PCR reaction with 5 µM each of TRV-specific primers (SP9238 + 9239) or NbPP2a-specific primers (SP8060 + 8061). PCR conditions included a denaturing step of 95 °C/3 min plus 40× (95 °C/5 s plus 60 °C/30 s). RT–PCR was performed on the remainder of the samples by QIAGEN OneStep RT–PCR using 50 ng of RNA, TRV-specific primers RN321 and RN322 and Ef1a control gene-specific primers AJ172 and AJ173. The PCR conditions consisted of 50 °C/30 min, 95 °C/15 min, 35×:(95 °C/30 s plus 58 °C/30 s plus 72 °C/1 min), 72 °C/10 min and 80 °C/5 min. About 10 µl of PCR products was analysed by agarose gel electrophoresis.

Reporting Summary. Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Data availability

High-throughput sequencing data have been deposited in the NCBI Sequence Read Archive database under BioProject accession no. PRJNA625335. TRV vectors are available from Addgene.

Received: 7 October 2019; Accepted: 22 April 2020;
Published online: 01 June 2020

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Acknowledgements

This work was supported by grant no. HR0011-17-2-0053 from the Defense Advanced Research Projects Agency (DARPA; to D.F.V. and S.D.-K.), funds from the Innovative Genomics Institute grant (to S.D.-K.) and National Science Foundation grant no. NSF-IOS-1339185 (to S.D.-K.). The University of Minnesota and University of California, Davis are part of a team supporting DARPA's Insect Allies Program under agreement no. HR0011-17-2-0053. We thank J.-Y. Liu for maintaining plants and for technical help during the revision of the manuscript. We thank M. Leffler for help with the figures.

Author contributions

E.E.E. designed the research, performed experiments with assistance from M.E.G., analysed data and wrote the manuscript. The work with tRNAs was carried out by U.N. and P.-J.H. D.F.V. and S.D.-K. supervised the research and helped write the manuscript.

Competing interests

E.E.E. and D.F.V. are named inventors on a patent application pertaining to technology filed by the University of Minnesota. D.F.V. serves as Chief Science Officer for Calyxt, an agricultural biotechnology company that uses gene editing to create new crop varieties.

Additional information

Extended data is available for this paper at https://doi.org/10.1038/s41477-020-0670-y.

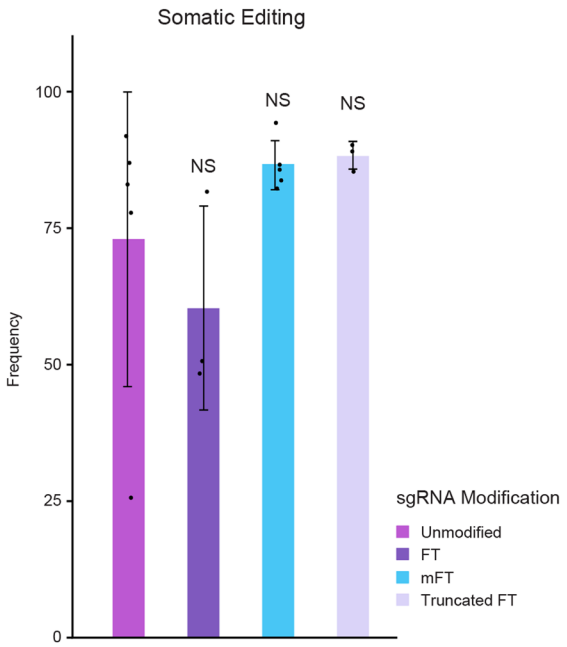
Supplementary information is available for this paper at https://doi.org/10.1038/s41477-020-0670-y.

Correspondence and requests for materials should be addressed to D.F.V.

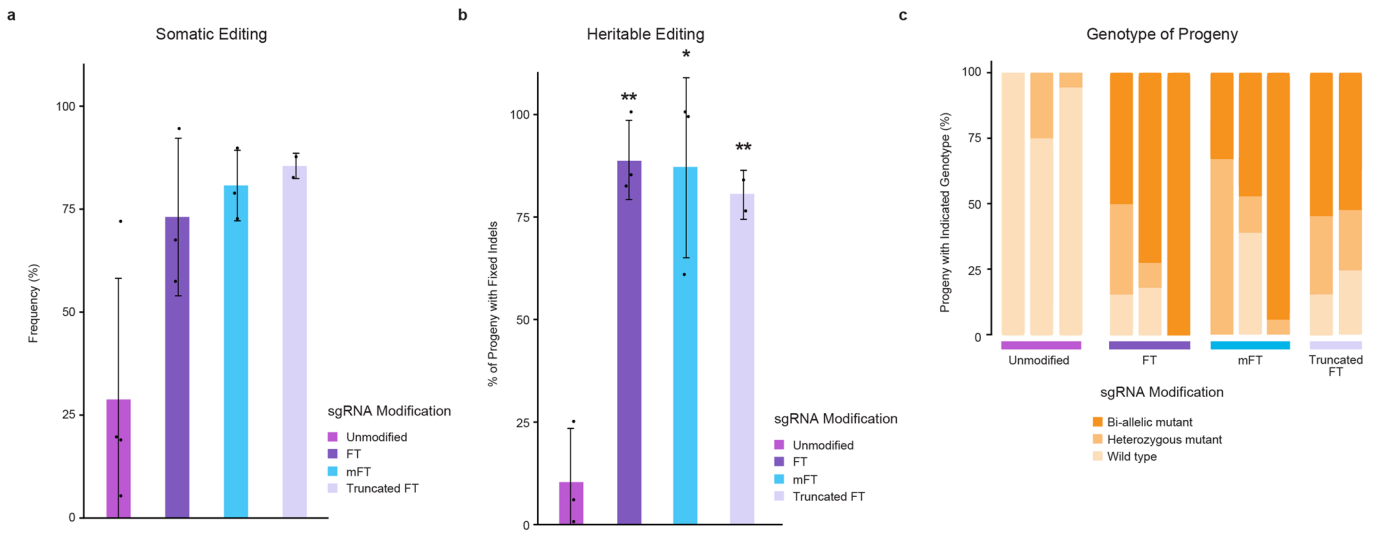
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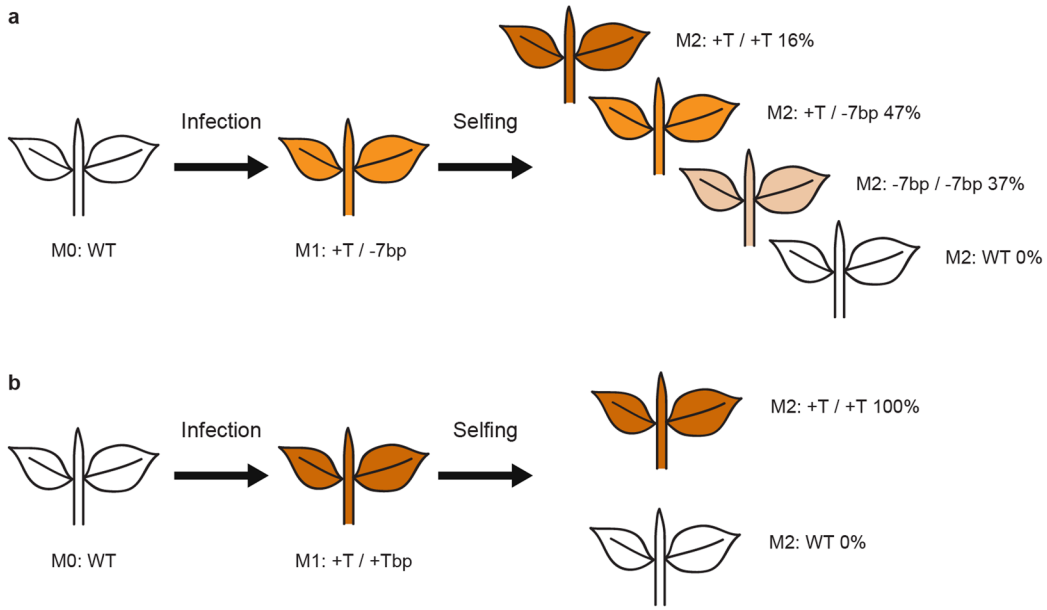
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Extended Data Fig. 1 | Addition of *Flowering Locus T* sequences to sgRNAs does not inhibit editing activity. There are no significant differences in editing efficiency when the sgRNA is either unmodified (n = 5 independent biological replicates, $\bar{x} = 0.73 \pm 0.27$) or modified with FT (n = 3, $\bar{x} = 0.61 \pm 0.19$, p-value= 0.477), mFT (n = 5, $\bar{x} = 0.87 \pm 0.05$, p-value= 0.328), or truncated FT (n = 3, $\bar{x} = 0.88 \pm 0.03$, p-value= 0.286). The modified or unmodified sgRNAs were expressed from TRV vectors, and tissue was collected from the infiltrated site in *N. benthamiana* leaves. Collecting tissue from the infiltrated site allows for examination of sgRNA function alone. DNA was extracted from the infiltrated site two weeks after infiltration. The targeted locus was PCR amplified and sequenced using Next Generation Sequencing (NGS). The Y axis denotes the fraction, displayed as a percentage, of reads with indels at the targeted locus divided by the total number of reads. Each dot represents one plant replicate. Bars indicate the mean editing frequency; error bars represent ± standard deviation. Asterisks indicate statistically significant differences compared with the unmodified sgRNA by the two-sided Student's t-test, NS > 0.05, *<0.05, **<0.01.



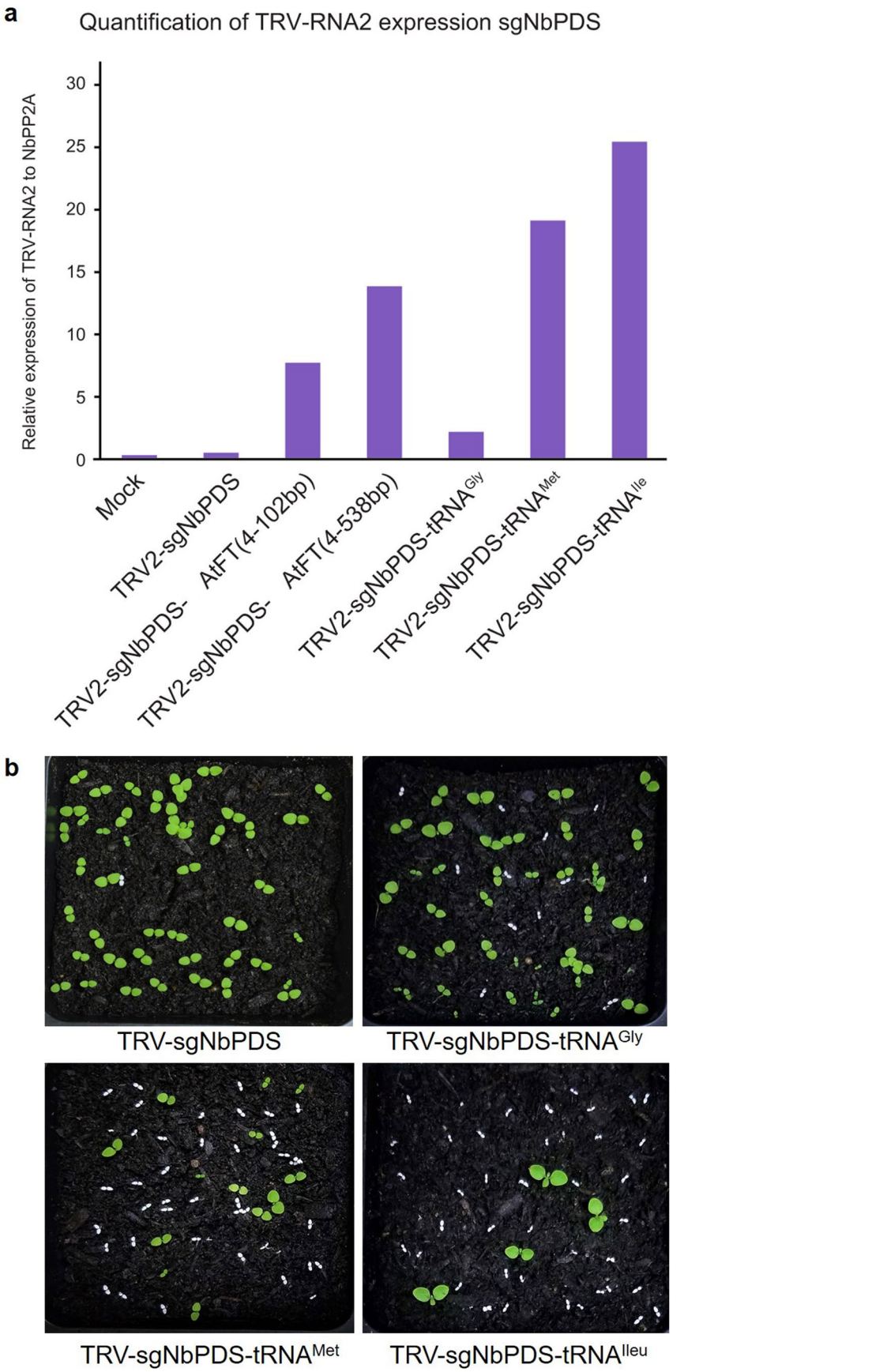
Extended Data Fig. 2 | Modified sgRNAs enable high frequencies of somatic and heritable editing at the *AG* locus. a, Editing efficiency of vectors targeting *AG* at the eighth systemic leaf. sgRNAs modified with FT (n = 3 independent biological replicates, $\bar{x} = 0.73 \pm 0.19$), mFT (n = 3, $\bar{x} = 0.81 \pm 0.09$), or Truncated FT (n = 2, $\bar{x} = 0.85 \pm 0.03$) yield higher average editing frequencies and lower variance compared to unmodified sgRNAs (n = 4, $\bar{x} = 0.29 \pm 0.29$). Tissue was collected approximately 6 weeks after infiltration. The Y axis is the percentage of NGS reads with indels divided by the total number of reads. Each dot represents one plant replicate. Bars indicate the mean editing frequency; error bars represent ± standard deviation. **b,** sgRNAs modified with FT (n = 3 independent biological replicates, $\bar{x} = 0.89 \pm 0.10$, p-value= 0.002) or mFT (n = 3, $\bar{x} = 0.87 \pm 0.22$, p-value= 0.011) or Truncated FT (n = 2, $\bar{x} = 0.81 \pm 0.06$, p-value= 0.004) produce significantly higher frequencies of heritable editing compared to unmodified sgRNAs (n = 3, $\bar{x} = 0.10 \pm 0.13$). In some cases, 100% of progeny contained a mutation. Heritable editing frequency is the fraction, displayed as a percentage, of progeny that contained an indel in at least one allele divided by the total number of progeny genotyped. Each dot represents the heritable editing frequency in progeny of one parent replicate infected with TRV expressing either unmodified sgRNAs (n = 13, 16, or 17) or modified with FT (n = 20, 11, or 14 progeny), mFT (n = 18, 21, or 19) or Truncated FT (n = 20 or 17). Bars indicate the mean editing frequency; error bars represent ± standard deviation. Asterisks indicate statistically significant differences compared to the unmodified sgRNA by the two-sided Student's t-test, *<0.05, **<0.01. (**c**) Modified sgRNAs produce a higher portion of bi-allelic and heterozygous mutations compared to unmodified sgRNAs. In one case, 100% of progeny contained bi-allelic mutations. The percentage of each zygosity was determined by the fraction of progeny that contained that genotype divided by total progeny assessed. The same progeny used to determine heritable editing frequency were used to calculate the genotype percentages.



Extended Data Fig. 3 | Inheritance of gene edits at the *N. benthamiana AG* locus. a, Genotypes of progeny from parents with observed +T/-7bp mutations. One seedling (M1) produced from a parent plant (M0) infected with TRV vectors expressing a sgRNA with Truncated FT had +T/-7bp mutations in *AG*. This plant was grown to maturity and seeds were collected from the first seed pod. Nineteen progeny of this plant were genotyped, and an approximate Mendelian inheritance was observed. **b,** Genotypes of progeny from parents with observed +T/+T mutations. One seedling (M1) from a parent plant (M0) infected with TRV vectors expressing a sgRNA with truncated FT had +T/+T mutations in *AG* alleles. This plant was grown to maturity and seeds were collected from the first seed pod. Nineteen progeny of this plant were genotyped, and the expected Mendelian inheritance was observed. Genotypes were determined through DNA extraction of each progeny, amplification of the *AG* locus, Sanger sequencing, and detection of indels by ICE (Synthego). Due to noise in ICE analyses, the following parameters were used to call genotype: 0%-10%=WT, 35%-65%= heterozygous (WT/indel), 85%-100%=bi-allelic mutation.



Extended Data Fig. 4 | Phenotypes of plants infected with TRV vectors expressing sgRNAs that target *N. benthamiana PDS*. Pictures of plants were taken between four and eight weeks after infection. The pictures display some of the typical phenotypes observed from either **a,** a non-infected plant or plants infected with a TRV vector expressing **b,** a non-augmented sgRNA, **c,** an FT-augmented sgRNA, **d,** a mFT-augmented sgRNA, **e,** a truncated FT-augmented sgRNA, **f,** a glycine tRNA-augmented sgRNA, **g,** a methionine tRNA-augmented sgRNA, **h,** an isoleucine tRNA-augmented sgRNA. Similar phenotypes were observed in three independent experiments, and each experiment consisted of 2–3 independent plants.

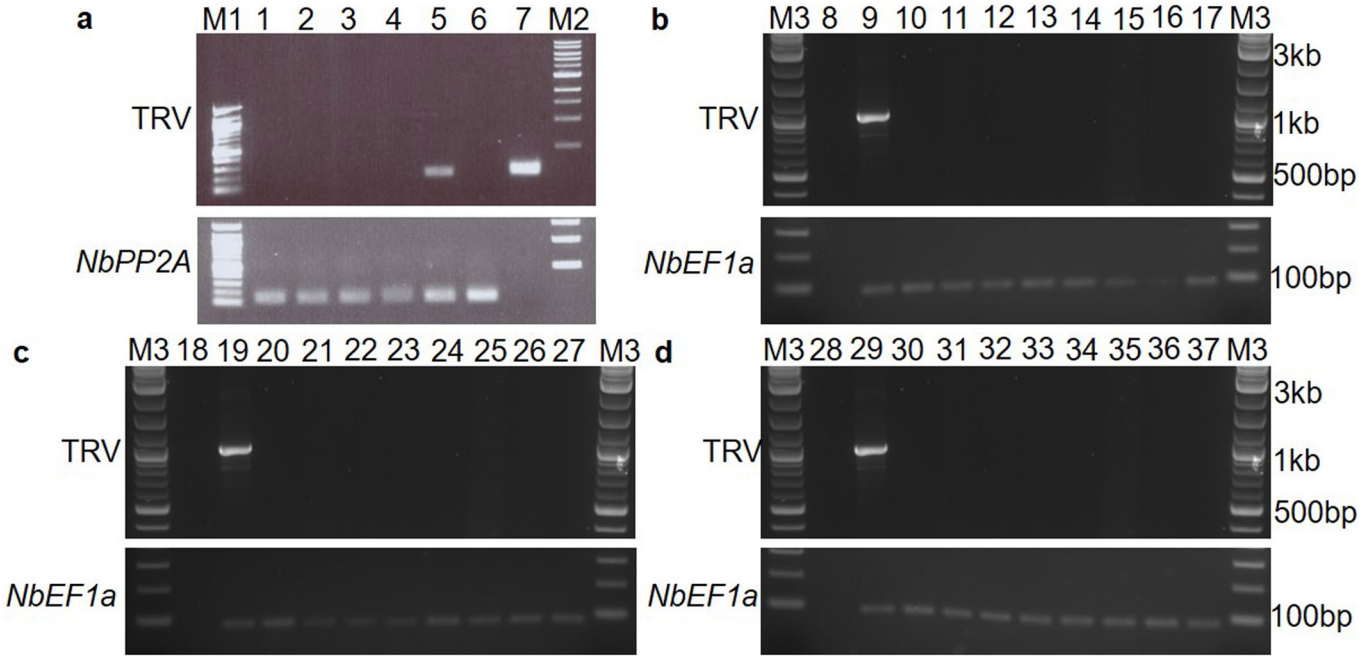


Extended Data Fig. 5 | See next page for caption.

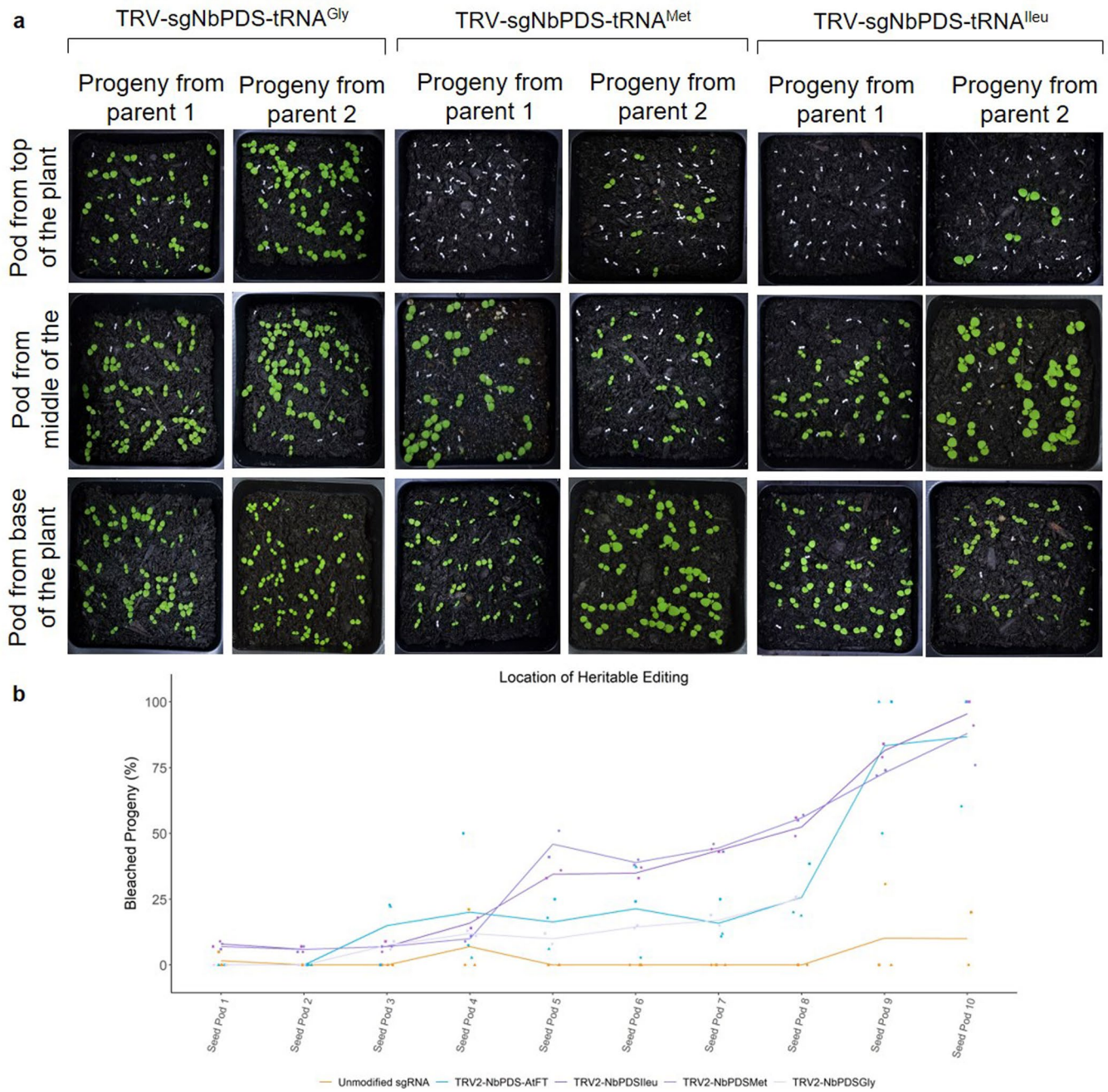
Extended Data Fig. 5 | TRV2 abundance reflects heritable editing efficiency. a, Relative TRV2 RNA levels were measured by quantitative real-time PCR relative to the control, *PP2A*. **b,** Green and bleached plant phenotypes of progeny from parental plants infected with a TRV2 vector expressing sgNbPDS, sgNbPDS-tRNA^{Ileu}, sgNbPDS-tRNA^{Met}, or sgNbPDS-tRNA^{Gly}. Frequencies of heritable editing correlate with viral abundance in the parent generation. About 50–60 progeny seeds from a pod from the top of the parent plants were planted on soil in individual pots. Pictures were taken 10–14 days post-seeding. This experiment was repeated twice with similar results.



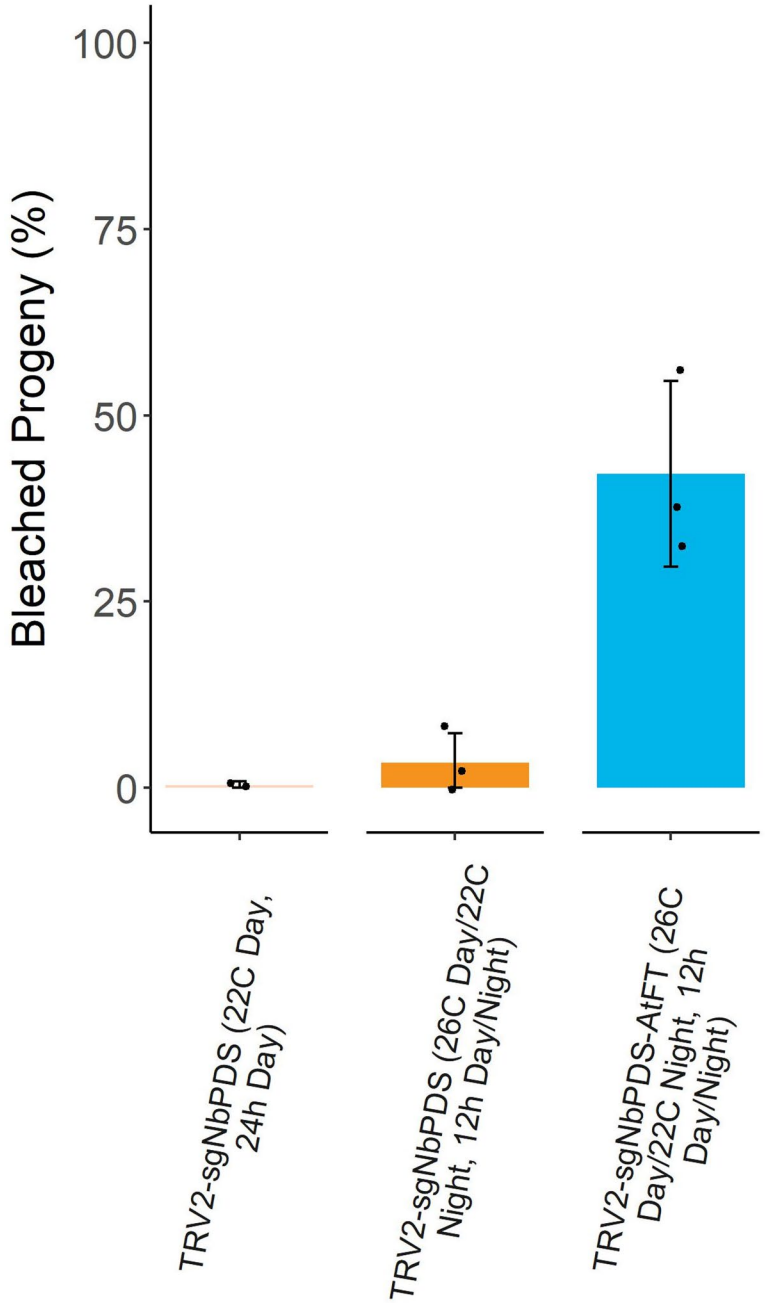
Extended Data Fig. 6 | Mutation types identified at the *NbPDS* locus in progeny plants derived from a parental plant infected with sgNbPDS-tRNA^{Ileu} or with sgNbPDS-tRNA^{Met}. DNA fragments flanking the *NbPDS* target site from progeny plants were cloned and 6–10 clones were sequenced. For each plant, the wild type sequence is shown at the top and the target site is shown in red and the protospacer adjacent motif (PAM) is underlined. Number of base addition (+) or deletion (−) are shown on the right.



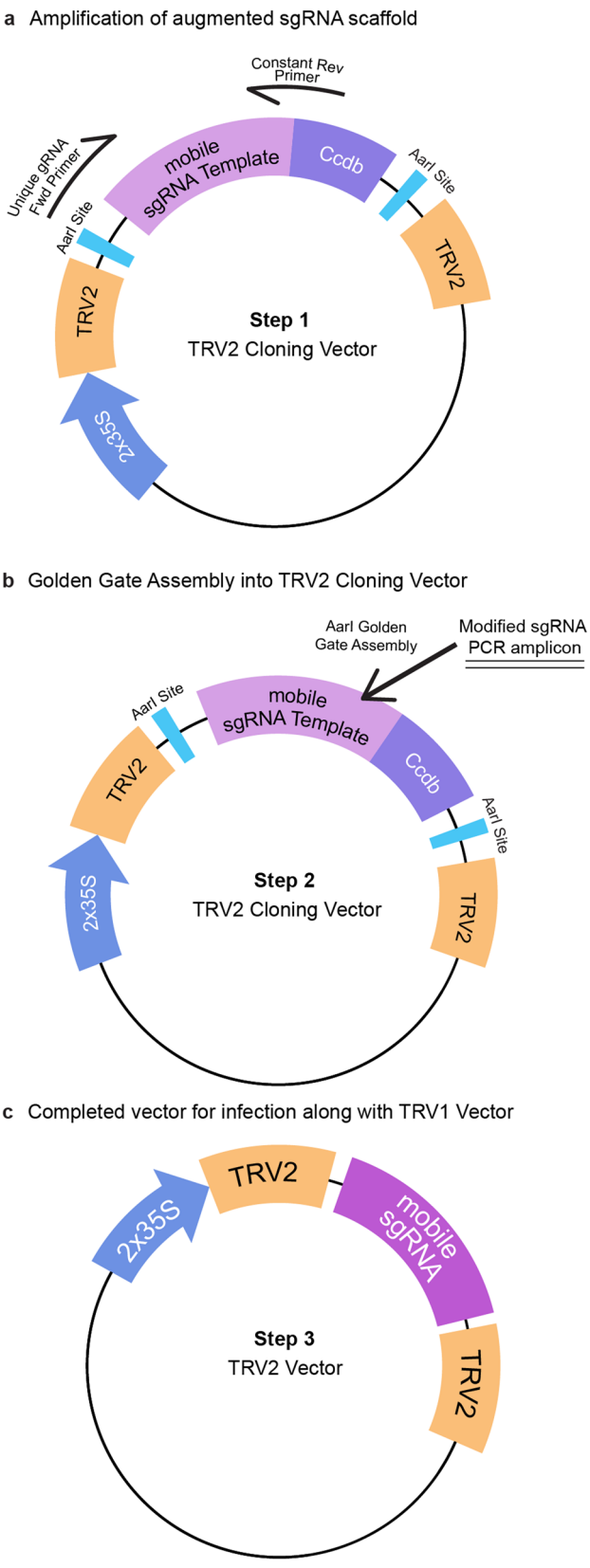
Extended Data Fig. 7 | TRV is not transmitted to the next generation. RT-PCR was performed using total RNA extracted from progeny heterozygous for mutations at *PDS* of plants infected with TRV editing vectors. **a,** Progeny RNA from a parental plant infected with TRV-sgNbPDS-tRNA^{Met} (lanes 1–2) and TRV-sgNbPDS-tRNA^{Ileu} (lanes 3–4), TRV-infected control (lane 5) and mock-infected control plant tissue (lane 6) using primers that amplify the TRV 3′-end (upper panel) and endogenous *NbPP2A* gene (bottom panel). Lane 7, TRV plasmid template used in PCR as a positive control. **b,** Progeny RNA from parental plant replicate 1 infected with TRV-sgNbPDS-FT (lanes 10–17), TRV-infected control (lanes 9, 19, and 29) and water control (lanes 8, 18, and 28) using primers that amplify TRV across the sgNbPDS-FT insert (upper panel) and endogenous *NbEF1a* gene (bottom panel). **c,** Progeny RNA from parental plant replicate 2 infected with TRV-sgNbPDS-FT (lanes 20–27). **d,** Progeny RNA from parental plant replicate 3 infected with TRV-sgNbPDS-FT (lanes 30–37). Lanes M1, 100 bp DNA ladder; M2, 1 kb DNA ladder; M3, NEB 1 kb+ ladder. Conclusions were the same for experiments performed at different institutions: the experiment in panel **a** was performed at UC Davis; the experiments in panels **b–d** were performed at the University of Minnesota.



Extended Data Fig. 8 | Heritable editing frequency increases as the infected plant matures. a, Green and bleached plant phenotypes of progeny from parental plants infected with a TRV vector expressing sgNbPDS-tRNA^{Ileu}, sgNbPDS-tRNA^{Met}, or sgNbPDS-tRNA^{Gly}. About 50–60 progeny seeds from a pod from the top of the parent plants (top panels), a pod from middle of the parent plants (middle panels) and a pod from base of the parent plants (bottom panels) were planted on soil in individual pots. Pictures were taken 10–14 days post-seeding. Similar phenotypes were observed in two independent experiments. **b,** Ratio of bleached progeny to total number of progeny from individual seed pods. Parental plants were infected with a TRV vector expressing sgNbPDS (Unmodified sgRNA, n = 3 independent biological replicates),sgNbPDS-Truncated FT (TRV2-NbPDS-AtFT, n = 3), sgNbPDS-tRNA^{Ileu} (n = 2), sgNbPDS-tRNA^{Met} (n = 2), or sgNbPDS-tRNA^{Gly} (n = 2). Seeds were collected from each seed pod individually with Seed Pod 1 being the first emerged seed pod and ascending vertically. Thirty to sixty progeny were phenotyped per pod. Three infected parent replicates were collected per treatment. Each shape corresponds to one infected parent replicate. Each line corresponds to the mean percentage of bleached progeny from each modified or unmodified sgRNA treatment.



Extended Data Fig. 9 | Growth conditions affect heritable editing frequency. Bleached progeny (%) indicates the ratio of bleached progeny to green progeny. Parental plants were infected with a TRV vector expressing sgNbPDS (TRV2-sgNbPDS) or sgNbPDS-Truncated FT (TRV2-sgNbPDS-AtFT). Progeny were collected from the entire infected parent plant and 150–300 seedlings were phenotyped. TRV2-sgNbPDS infected plants were grown at two different conditions: 22 °C with 24 h light (n = 2 independent biological replicates, $\bar{x} = 0.004 \pm 0.005$) or 26 °C day/22 °C night with a 12 hour day/night cycle (n = 3, $\bar{x} = 0.034 \pm 0.039$). All TRV2-sgNbPDS-AtFT were grown at 26 °C day/22 °C night with a 12 hour day/night cycle (n = 3, $\bar{x} = 0.421 \pm 0.125$). Each dot represents one plant replicate. Bars indicate the mean editing frequency; error bars represent ± standard deviation.



Extended Data Fig. 10 | Method for rapid cloning of augmented sgRNAs into TRV vectors. A cloning vector was developed that allows for simple and efficient cloning of unique sgRNA target sequences into TRV2 vectors. Unique sgRNAs are added to the vector using one unique primer (that contains the sgRNA target of interest) and one reverse primer (that can be used for all unique primers). These primers are used to amplify the modified sgRNA from the TRV2 cloning vector backbone. This amplicon is then added to the TRV2 cloning vector through a Golden Gate Assembly cloning step to create a final TRV2 vector with the modified sgRNA of interest.

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Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	No software was used for data collection.
Data analysis	All software used for DNA sequence analysis is cited in the manuscript and methods. Specifically, ICE v2.0 (Synthego) was used to quantify indel frequencies; CRISPR RGEN Tools was used to analyze NGS data.

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Life sciences study design

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Sample size	The somatic cell editing sample size included all plants infected with viral vectors expressing a given mobile sgRNA. The heritable editing sample size was chosen by collecting seeds from all infected parents. Progeny sample size captureed editing efficiency of at least 10% (at least 10 progeny were genotyped from each infected parent). All seeds from infected parents were pooled before genotyping to capture the spectrum mutations across the whole plant.
Data exclusions	No data was excluded.
Replication	Heritable editing was demonstrated with high frequency at three genomic loci in multiple experiments and plant replicates. Similar editing frequencies and phenotypes were also observed in experiments performed at separate institutions, the University of Minnesota and University of California, by separate researchers.
Randomization	Experimental groups were defined by the specific mobile sgRNA that was delivered to plants via viral vector (including controls). Randomization was not relevant to the study as all plants were grown under the same conditions.
Blinding	Tissue was extracted from the same location for all somatic assays. All seeds from each infected parent were collected together and planted indiscriminately. Seedlings resulting from PDS targeting experiments were genotyped corresponding to the ratio of bleached to green seedlings (ex. one parent produced 23 bleached and 58 green seedlings meaning 5 bleached and 12 green seedlings were genotyped). AG targeting experiments produced no phenotype and thus blinding was not relevant.

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Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging